

Civil Engineering & Structural Design — CodeAlpha

Civil Engineering & Structural Design — Internship Overview

This internship program provides practical exposure to civil engineering principles and structural design techniques. CodeAlpha is dedicated to building industry-ready professionals skilled in designing safe, reliable, and sustainable structures. The internship equips students with knowledge of building design, structural analysis, construction materials and drafting tools such as AutoCAD, STAAD.Pro or ETABS. Interns will gain hands-on experience in preparing 2D drawings, analyzing load-bearing structures and working on mini design projects. Through expert mentorship and real-world assignments, participants will develop technical and analytical skills required for careers in construction, infrastructure, and structural engineering.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
CodeAlpha_ProjectName
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Civil Engineering & Structural Design Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Residential Building Floor Plan

Design a **2D floor plan of a G+1 residential building** using AutoCAD.

- Include dimensions, wall thickness, and furniture layout.
 - Add doors, windows, and labeling.
 - 👉 Submit CAD file + PDF export.
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✓ TASK 2: Structural Load Analysis

Perform a **load calculation and analysis** for a simple structure (beam, slab, or column).

- Consider **dead load, live load and wind/earthquake load (basic level)**.
 - Present calculations and structural behavior.
👉 Submit report (2–3 pages) + calculation sheet.
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✓ TASK 3: 3D Modeling & Visualization

Create a **3D model of a small structure** (house, bridge component, or commercial block) using AutoCAD/Revit/SketchUp.

- Apply materials and rendering for visualization.
👉 Submit design file + rendered images.
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✓ TASK 4: Mini Project – Structural Design Proposal

Prepare a **mini project report (5–6 pages)** on one of the following:

- Design & analysis of a small bridge
 - Sustainable building design using green materials
 - Earthquake-resistant building design concepts
 - High-rise building load distribution study
👉 Submit report in PDF with diagrams/charts.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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